

Simulation and Comparative Study of FOC Control Algorithms for Permanent Magnet Synchronous Motors

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ABSTRACT

This paper focuses on the Field-Oriented Control (FOC) technology of Permanent Magnet Synchronous Motors (PMSM). On the MATLAB/Simulink simulation platform, the performance of traditional PI control, Sliding Mode Control (SMC), and Extended Kalman Filter (EKF) algorithms are discussed and compared. By establishing a complete mathematical model and control strategy, the advantages and disadvantages of each algorithm in speed response, torque ripple, and anti-interference ability are analyzed. Simulation results show that traditional PI control has good stability, SMC has excellent robustness, and EKF can effectively achieve high-precision sensorless estimation. This study provides a theoretical basis and simulation reference for the selection of PMSM control schemes in different application scenarios.

KEYWORDS

PMSM; FOC; PI Control; SMC; EKF

1 Introduction

With the development of permanent magnet materials, motor manufacturing, and FOC algorithms, PMSM has been widely used in many fields. Due to small size, light weight, and high power density, PMSM is suitable for many industrial applications.

In recent years, advanced algorithms such as neural networks, sliding mode control, Kalman filtering, and PID control have improved the anti-interference and robustness of PMSM drives ^[1]. Hong et al. studied fuzzy sliding mode control for PMSM ^[2]. Wu et al. proposed a sensorless FOC method based on EKF ^[3].

However, there is a lack of systematic comparison among these algorithms. This paper builds and compares different control algorithms on MATLAB/Simulink to clarify their applicable scenarios.

2 Principle Analysis

FOC was first proposed by Blaschke in 1971 ^[4]. It uses coordinate transformations including Clark, Park, inverse Park, and SVPWM to achieve decoupled control of PMSM. The control method is based on the principle of coordinate transformation. Its core modules include Clark transformation, Park transformation, d/q axis current proportional control, inverse Park transformation, and Space Vector Pulse Width Modulation (SVPWM) modules. Through Clark transformation and Park transformation, the stator current of the motor can be decoupled into independent excitation component i_d and torque component i_q . Then, the two components are separately closed-loop regulated by the d/q axis current PI controller, which makes the AC motor with vector control equivalent to the DC motor with scalar control, greatly simplifying the complex mathematical model of strong coupling of AC motors. This control method was initially applied to asynchronous motors, and later extended to Brushless DC Motors (BLDCM) and PMSM control systems, both achieving excellent speed regulation dynamic performance. Below, I will introduce and explain the core technologies in the FOC algorithm.

2.1 Clark Transformation

Clark transformation projects the current, voltage, and flux linkage on the A, B, and C three-phase windings of the motor stator onto a two-phase stationary orthogonal coordinate system where the A-phase axis is the α axis and the axis perpendicular to the α axis and leading by 90° electrical angle is the β axis. Using the symmetry of the three-phase windings, the three-phase is reduced to two-phase, eliminating the coupling relationship between the three-phase quantities, simplifying the control model, and preparing for the subsequent Park transformation.

$$\begin{cases} i_\alpha = \sqrt{\frac{2}{3}} \left(i_a - \frac{1}{2}i_b - \frac{1}{2}i_c \right) \\ i_\beta = \sqrt{\frac{2}{3}} \left(\frac{\sqrt{3}}{2}i_b - \frac{\sqrt{3}}{2}i_c \right) \end{cases}$$

2.2 Park Transformation

It projects the AC quantity in the $\alpha\beta$ two-phase stationary coordinate system onto the dq orthogonal coordinate system rotating synchronously with the motor rotor. It converts the alternating $\alpha\beta$ axis components into constant dq axis components during the steady-state operation of the motor, realizing the complete decoupling of the excitation current i_d and the torque current i_q , and making the control of the AC motor equivalent to the independent control of the armature and excitation windings of the DC motor, which is the core step of the FOC algorithm. The key input of the transformation is the rotor position angle θ , which needs to be collected in real-time through sensors or sensorless algorithms.

$$\begin{cases} i_d = i_\alpha \cos \theta - i_\beta \sin \theta \\ i_q = i_\alpha \sin \theta + i_\beta \cos \theta \end{cases}$$

2.3 Inverse Park Transformation

The function of the inverse Park transformation is to convert the DC quantity in the two-phase rotating dq coordinate system back to the AC quantity in the two-phase stationary $\alpha\beta$ coordinate system. In FOC control, the output of the PI regulator is the dq axis voltage control quantity, which must be converted back to the AC quantity in the stationary coordinate system through the inverse Park transformation to provide input signals for subsequent SVPWM modulation.

$$\begin{cases} u_\alpha = u_d \cos \theta - u_q \sin \theta \\ u_\beta = u_d \sin \theta + u_q \cos \theta \end{cases}$$

2.4 Inverse Clark Transformation

The function of the inverse Clark transformation is to restore the AC quantity in the two-phase stationary $\alpha\beta$ coordinate system to the three-phase physical quantity in the three-phase stationary abc coordinate system. The $\alpha\beta$ axis voltage obtained through the inverse Park transformation needs to be remapped to the voltage of the motor stator three-phase windings through the inverse Clark transformation, and finally output to the inverter to drive the motor to run.

$$\begin{cases} u_a = \sqrt{\frac{2}{3}} u_\alpha \\ u_b = \sqrt{\frac{2}{3}} \left(-\frac{1}{2} u_\alpha + \frac{\sqrt{3}}{2} u_\beta \right) \\ u_c = \sqrt{\frac{2}{3}} \left(-\frac{1}{2} u_\alpha - \frac{\sqrt{3}}{2} u_\beta \right) \end{cases}$$

2.5 SVPWM

Space Vector Pulse Width Modulation (SVPWM) technology generates an approximately circular magnetic field by synthesizing space voltage vectors to control the inverter^[5]. Among them, the average value equivalent principle aims to keep the average value of the voltage vector within the control cycle consistent with the expected average value of the voltage. The target voltage vector of SVPWM can be obtained by adding adjacent basic voltage vectors. Combined with the area equivalent and PWM principles, the voltage component size can be determined through the voltage action time, and vice versa; by adjusting the on-time of the power switch tubes of the inverter bridge, the PWM voltage generated by different switch tubes is used to synthesize a voltage vector that moves circularly in space, thereby generating a circularly moving flux linkage, and finally obtaining the circular magnetic field required for motor operation. In addition, SVPWM adjusts the space voltage vector by controlling the switching time and frequency, and optimizes the symmetry of the PWM signal by regulating the action time of the zero vector, effectively reducing the current harmonic component and improving the power quality of the system; the optimization of current harmonics can reduce switching loss and improve power utilization.

3 Simulation Model Introduction

This section introduces the basic FOC simulation model based on the $i_d = 0$ magnetic field orientation strategy. First, the speed of this model and the subsequent models will be set to 500 rpm, and a sensor will be connected for the deviation calculation link, which will be subtracted from the actual speed signal fed back by the motor to obtain the speed deviation. Secondly, the model adopts a relatively simple control strategy of $i_d = 0$, because at this time the electromagnetic torque of the motor is only proportional to the q-axis current, and the d-axis no longer participates in torque generation. Torque adjustment can be achieved only by controlling i_q , and the d and q-axis controls are

independent of each other, without cross-coupling interference and no need to add additional decoupling compensation terms. Then, combined with the q-axis current command, the deviation calculation is performed with the motor feedback current signals of the d and q axes respectively to obtain the deviations of the two current signals, and then the d and q axis voltages in the rotating coordinate system are decoupled.

After decoupling, the dq axis voltage and the real angle of the motor rotor are jointly input into the inverse Park transformation module to complete the task from the dq coordinate system to the $\alpha\beta$ coordinate system. Then the voltage signal is input into the SVPWM pulse modulation module, which converts the target voltage vector into the corresponding pulse signal of the power tube of the three-phase inverter bridge, drives the three-phase full-bridge inverter powered by the DC bus to work, and divides the DC power into three-phase AC power with adjustable amplitude and phase, thus completing the driving of the motor. A step load module is also added to the model. The TL module in the figure is connected to TM to simulate the load. The parameter is set to start from 0 with a load of 5N/m, which can better simulate the real working scene, provide external load torque disturbance for the motor, and is also suitable for the subsequent analysis of the system's anti-interference performance.

In the closed-loop feedback, the three-phase stator current of the motor passes through Clark co- ordinate transformation and Park coordinate transformation in turn, and combines with the real-time electrical angle of the rotor to decouple the d and q axis current signals, which are fed back to the current deviation calculation link to complete the current closed-loop feedback; at the same time, the real electrical angle of the rotor participates in the operation of each coordinate transformation module throughout the process to achieve precise field-oriented control. The simulation environment is uniformly configured by the powergui module. The entire system adopts discrete fixed-step simulation, and the simulation step size is set to $1\mu\text{s}$, which is consistent with the control timing of the actual digital controller, ensuring the simulation operation accuracy and dynamic characteristics. On the basis of the above, I will add a variety of control modules in the following to improve the control accuracy of the motor and the robustness of the model, and compare the simulation results of each model below.

The modeling method and simulation framework are established according to the classic PMSM modeling and simulation method proposed in the book of Yuan [7].

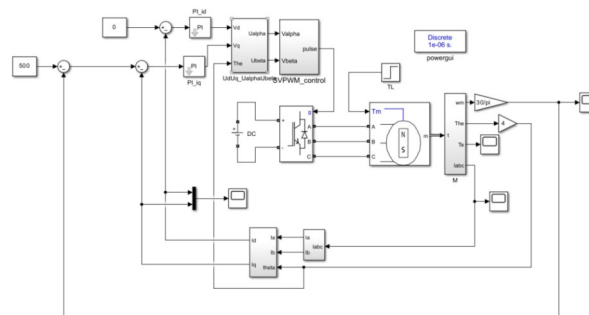


Figure 1 Basic FOC Simulation Model

4 PI Control

In the following permanent magnet synchronous motor vector control system, PI control (Proportional- Integral Control) is the core algorithm to realize the closed-loop regulation of speed and current. Through the proportional operation and integral operation of the error signal, the motor speed and current can quickly track the given value, and ensure no steady-state error. The input of the PI controller is the deviation between the given value and the actual feedback value, and the output is the control quantity. Its control law can be expressed as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau$$

In the formula: $e(t)$ is the control deviation, such as the speed deviation $\omega^* - \omega$ or the current deviation $i^* - i$; K_p is the proportional coefficient, which is responsible for quickly responding to the deviation and improving the system response speed; K_i is the integral coefficient, which is responsible for accumulating the deviation and eliminating the steady-state error. In the double closed-loop vector control structure of the motor, PI control is divided into two layers: 1. Current loop PI controller: mainly used to track the d and q axis current commands and suppress current disturbances. 2. Speed loop PI controller: mainly used to stabilize the motor speed. When the load changes suddenly, the proportional term quickly adjusts the output torque current, and the integral term accumulates the static error. In a word, PI control has the advantages of simple structure, easy parameter tuning and high reliability, which can meet the basic requirements of the system.

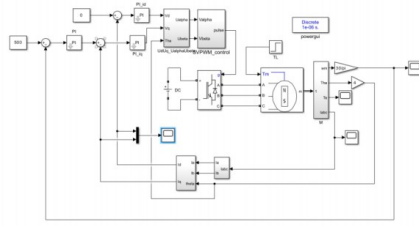


Figure 2 PI Module

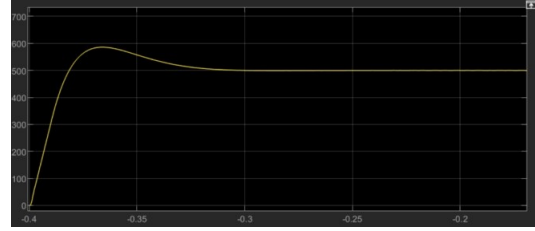


Figure 3 Speed Response (PI)

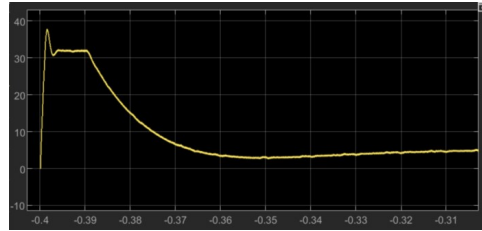


Figure 4 Torque Response (PI)

5 Sliding Mode Control (SMC)

Sliding Mode Control (SMC for short) does not require an accurate mathematical model of the controlled object, and has the advantages of fast response and insensitivity to parameter changes and disturbances ^[6]. The disadvantage is that the discontinuous control method causes large chattering. SMC is a nonlinear robust control algorithm, whose principle is to design the sliding mode surface and control law. In the system, SMC is used in the speed loop to improve the system robustness. First, define the speed error $e = \omega^* - \omega$. Design the linear sliding mode surface $s = ce + e$ ($c > 0$). Adopt the constant speed reaching law $\dot{s} = -k \cdot \text{sgn}(s)$ ($k > 0$). The SMC control law is derived:

$$u = \frac{1}{g} \left(-f + \dot{\omega}^* - c\dot{e} - k \cdot \text{sgn}(s) \right)$$

To suppress chattering, a saturation function is adopted:

$$\text{sat}\left(\frac{s}{\phi}\right) = \begin{cases} 1, & s > \phi \\ \frac{s}{\phi}, & |s| \leq \phi \\ -1, & s < -\phi \end{cases}$$

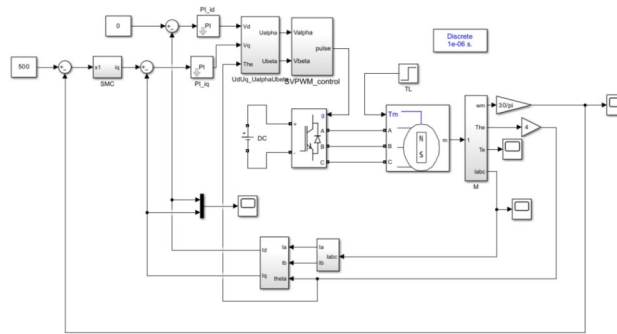


Figure 5 SMC Simulation Model

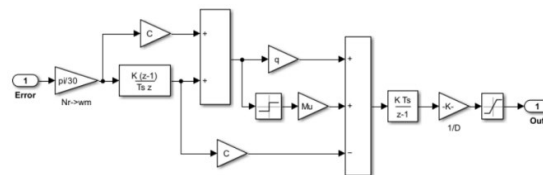


Figure 6 SMC Module

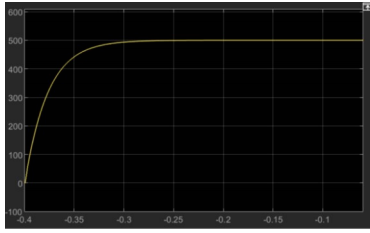


Figure 7 Speed Response (SMC)

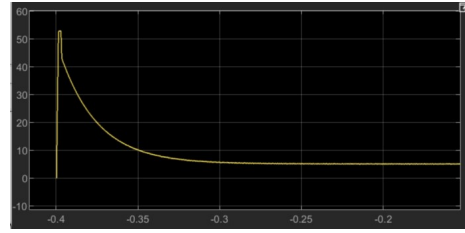


Figure 8 Torque Response (SMC)

6 Extended Kalman Filter (EKF)

Extended Kalman Filter (EKF) is an optimal state estimation algorithm for nonlinear systems with faster response and more stable waveform output. In the system, EKF is used to estimate the back electromotive force, rotor position and speed. After adding EKF, the response is faster, the redundant system interference is filtered out, and the feedback is more rapid. Its principle is to establish the motor system state prediction $x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}$ and the observation equation $z_k = h(x_k) + v_k$. Linearize the nonlinear function. High-precision estimation is realized through the cyclic iteration of prediction and correction.

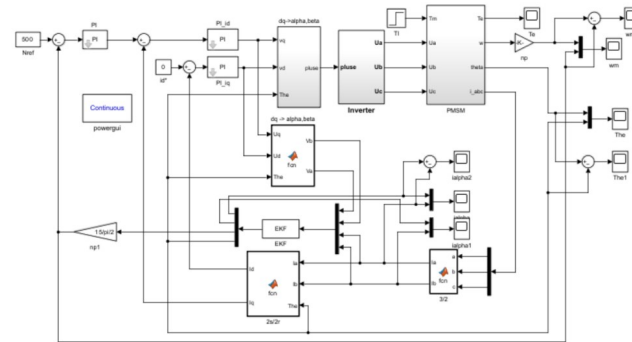


Figure 9 EKF Simulation Model

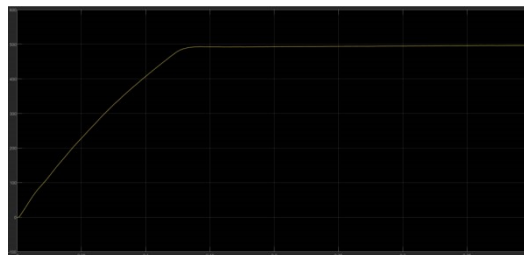


Figure 10 Speed Response (EKF)

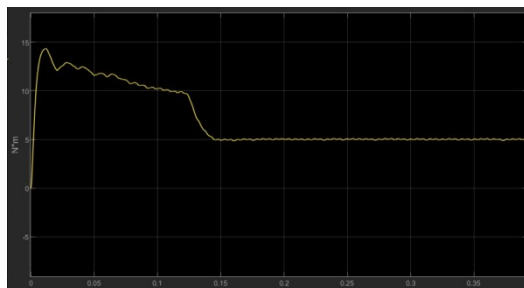


Figure 11 Torque Response (EKF)

7 Results and Conclusion

7.1 Results Analysis

Based on the above simulation results, it can be seen that the three control schemes of basic PI-FOC, sliding mode SMC-FOC and EKF sensorless FOC have obvious differences in speed regulation performance, anti-interference ability, torque ripple and implementation complexity. Among them, the basic PI-FOC has a simple structure, easy parameter tuning, smooth torque waveform without chattering, good steady-state accuracy and operational stability, but weak anti-load disturbance ability and dependence on physical position sensors; the sliding mode SMC-FOC has faster dynamic

response, stronger anti-interference ability and robustness, small speed fluctuation and fast recovery speed under sudden load changes, which can effectively suppress parameter perturbation and external disturbance, but there is obvious torque chattering due to the control principle, and the algorithm implementation and parameter debugging are relatively complex; the EKF sensorless FOC can realize sensorless operation through state observation. After omitting the position sensor, the system structure is simpler, the cost is lower, and the environmental adaptability is stronger. The speed estimation accuracy is higher in the medium and high speed range. However, affected by observation noise and model error, the torque smoothness is slightly lower than that of traditional PI control, and the algorithm has a larger amount of calculation, which has higher requirements on the hardware computing capacity. Each of the three algorithms has its own advantages and disadvantages, and the applicable scheme can be selected according to the speed regulation accuracy, robustness requirements, cost constraints and installation conditions, providing a reference for the design of permanent magnet synchronous motor control systems.

7.2 Conclusion and Prospect

This paper simulates, analyzes and compares the three control schemes of PI-FOC, sliding mode SMC-FOC and EKF sensorless FOC of permanent magnet synchronous motors, and also divides the advantage ranges. Each of the three algorithms has its own advantages and disadvantages, which can provide a reference basis for the selection of motor speed regulation system. However, there are still many deficiencies, such as a small number of analyzed samples, and no actual on-site construction and data measurement of these models. In addition, some new models and optimizations of old models have been developed during the writing of the article. There are many aspects of future optimization: first, improve the sliding mode and observer algorithms to suppress chattering and improve low-speed estimation accuracy; second, introduce intelligent algorithms such as neural networks, fuzzy control and deep learning to realize parameter self-tuning and nonlinear adaptive optimization; third, study advanced methods such as model predictive control MPC; fourth, build a composite control architecture integrating multiple algorithms, and finally transplant the algorithm to the hardware platform for experimental verification to promote engineering application.

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